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Instruction

What the question is

Context Free Grammars & Context Free Languages

CFG Parsing: 2.1

2.1 Recall the CFG G_4 that we gave in Example 2.4. For convenience, let's rename its variables with single letters as follows.

$$E \rightarrow E + T \mid T$$

$$T \rightarrow T \times F \mid F$$

$$F \rightarrow (E) \mid a$$

Give parse trees and derivations for each string.

a. a

c. $a+a+a$

b. $a+a$

d. $((a))$

2.1 D.C = Double Check

a) a. $E \rightarrow E + T \mid T \quad T \rightarrow T * F \mid F \quad F \rightarrow (E) \mid a \quad E - T - F - A$

Deriv. a $E \Rightarrow T$

$E \Rightarrow F$

$E \Rightarrow a$

b) $a + a$

$$\frac{E \vdash T \vdash F \vdash a}{E \vdash E \vdash T \vdash F \vdash a} + 4PT$$

Deriv: $a \circ a$

$E \rightarrow E + T$

$E \rightarrow A + T$

$E \rightarrow F + T$

$E \rightarrow a \circ T$

$E \rightarrow a + F$

$E \rightarrow a \circ a$

c) $a + a + a$

$E \begin{cases} T - F - a \\ \text{---} + \\ E \begin{cases} T - F - a \\ \text{---} + \\ E - T - F - a \end{cases} \end{cases}$

Deriv: $a + a + a$

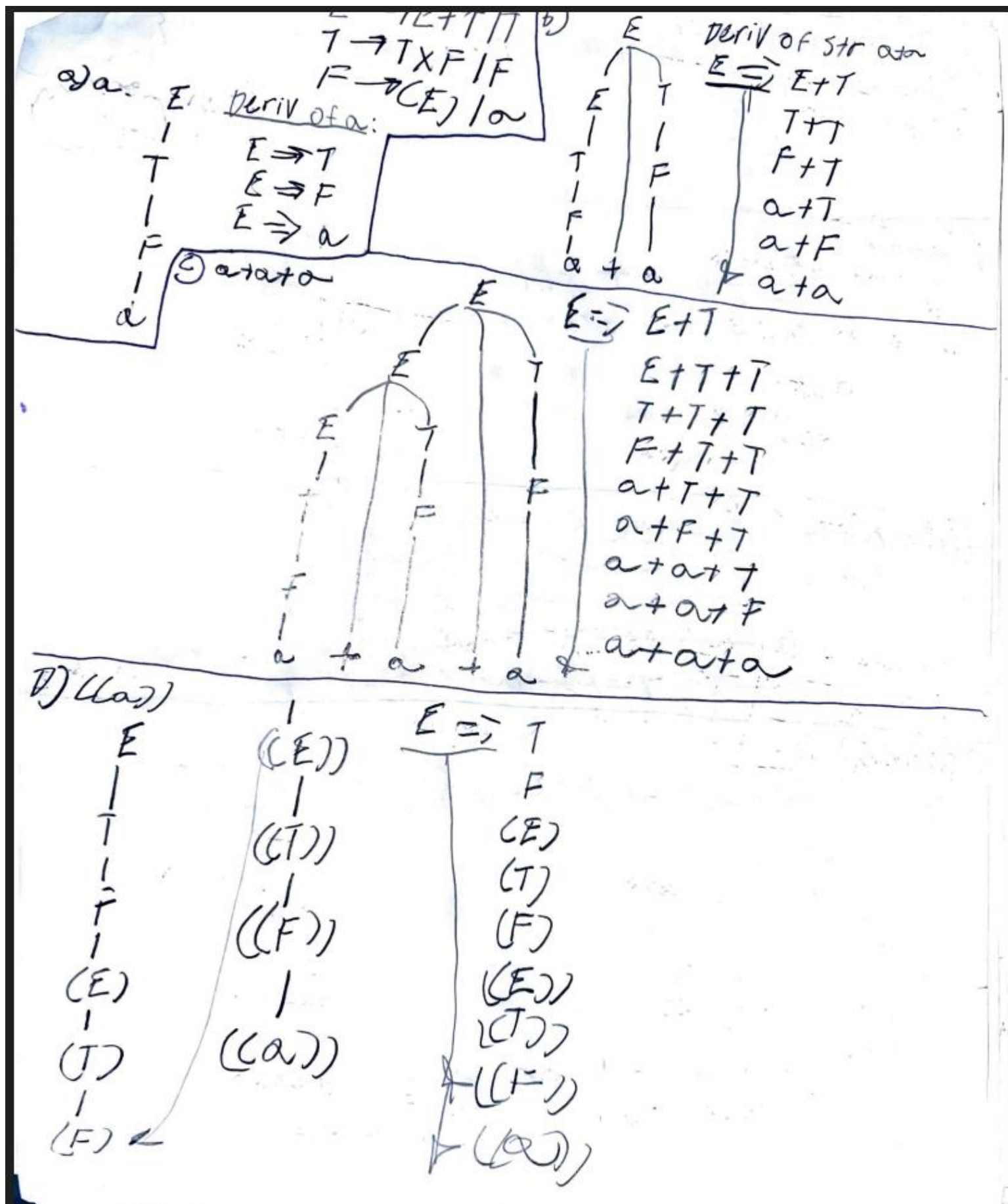
$E \rightarrow E + T$
 $E \rightarrow E * T + T$
 $E \rightarrow T + T + T$
 $E \rightarrow F + T + T$
 $E \rightarrow a + T + T$

d) $(2a) E-T$

$$E - T - F - (E) - (T) - (F) - ((E)) - ((T)) - ((F)) - ((a))$$

Deriv: (a)

$$E \rightarrow T$$
$$E \rightarrow F$$
$$E \rightarrow (E)$$
$$E \rightarrow CT$$
$$E \rightarrow F$$
$$E \rightarrow \mathcal{U}(E))$$
$$E \rightarrow (C \uparrow)$$
$$E \rightarrow (EF)$$
$$E \rightarrow (L \wedge \neg L)$$



^A2.3 Answer each part for the following context-free grammar G .

$$\begin{aligned}R &\rightarrow XRX \mid S \\S &\rightarrow \mathbf{a}T\mathbf{b} \mid \mathbf{b}T\mathbf{a} \\T &\rightarrow XTX \mid X \mid \epsilon \\X &\rightarrow \mathbf{a} \mid \mathbf{b}\end{aligned}$$

- a. What are the variables of G ?
- b. What are the terminals of G ?
- c. Which is the start variable of G ?
- d. Give three strings in $L(G)$.
- e. Give three strings *not* in $L(G)$.
- f. True or False: $T \Rightarrow \mathbf{aba}$.
- g. True or False: $T \xRightarrow{*} \mathbf{aba}$.
- h. True or False: $T \Rightarrow T$.
- i. True or False: $T \xRightarrow{*} T$.
- j. True or False: $XXX \xRightarrow{*} \mathbf{aba}$.
- k. True or False: $X \xRightarrow{*} \mathbf{aba}$.
- l. True or False: $T \xRightarrow{*} XX$.
- m. True or False: $T \xRightarrow{*} XXX$.
- n. True or False: $S \xRightarrow{*} \epsilon$.
- o. Give a description in English of $L(G)$.

2.3)

Sub = Substitute because = b/c
 F = False S = string
 T = True in c) - 0)

- a) variables G RSTX & Nonterminal sym
- b) Term G = aya & Terminal sym
- c) Start var $G = R$ $R \rightarrow XRX$ IS start variable = left hand side on the top

d) so $R \rightarrow S$ substitute $S \rightarrow aTb$
 $R \rightarrow aTb \rightarrow T \rightarrow \epsilon$
 $R \rightarrow ab$

Case 1

so $R \rightarrow S$ sub $S \rightarrow Ta$ $R \rightarrow bTa$ sub $T \rightarrow \epsilon$
 $R \rightarrow b\epsilon a$ $R \rightarrow ba$
 remove

Case 2

so $R \rightarrow S$ sub $S \rightarrow aTb$
 $R \rightarrow aTb$
 sub $T \rightarrow X$ $R \rightarrow aXb$ sub $X \rightarrow a$
 $R \rightarrow aab$

Case 3

Thus $L(G)$ ab, ba & aab

e) The 3 strings that CANNOT be in S cannot be deriv from grammar G = aba, b & ϵ b/c these

- f) F b/c the S is not deriv. using G
- g) T b/c the S can be deriv. using G
- h) F S cannot deriv
- i) T S can be deriv
- j) T S can be deriv
- k) E S cannot deriv
- m) T can be D
- n) F cannot be

$T \rightarrow aba$
 $T \rightarrow aba$
 $T \rightarrow T$
 $T \rightarrow T$
 $XXX \rightarrow aba$
 $X \rightarrow aba$
 $T \rightarrow XXX$
 $S \rightarrow \epsilon$

o) $L(G)$ has all S that AREN'T Palindromes formed terminal sym a & b

2.3 D.C? (variables = var)

(A I T R O G = appear in the rules of Grammar)

a. The var of G = R, S, T & X Var are non-terminal symbols

that A I T R O G

b. The Terminals of G are a, b. Terminals are the term sym

that A I T R O G

c. Start Var of G = R

R $\rightarrow$ XRXIS start Var. is var usually occurs on the left.
 hard side of the topmost Rule.

d) R $\rightarrow$ S Substitute = Sub Sub S $\rightarrow$ aTb so R $\rightarrow$ aTb

Sub T $\rightarrow$ E (R $\rightarrow$ aEb R $\rightarrow$ aTb case 2: R $\rightarrow$ S & Rule

Sub S w/ S $\rightarrow$ bTA R $\rightarrow$ bTA Sub T w/ T $\rightarrow$ E

R $\rightarrow$ bEa R $\rightarrow$ bEa Case 3: R $\rightarrow$ S & Rule Sub S w/ S $\rightarrow$ aTb

R $\rightarrow$ aTb Sub T $\rightarrow$ X R $\rightarrow$ aXb Sub X w/ Rule X $\rightarrow$ a R $\rightarrow$ aTb

Thus, 3 str L(G) = aTb, bEa, E

e. 3 str not in L(G) = aTb & E. Since These str cannot be derived from given grammar G

f. F = False

T $\rightarrow$ aTb, str cannot be deriv w/G.

g) T

T $\rightarrow$ aTb, str can be deriv w/G

h) F T $\rightarrow$ TT, str cannot be deriv w/G.

i) T T $\rightarrow$ T, str can be deriv w/G

j) T

xxx $\rightarrow$ aTb, str can be deriv w/G

k) F

x $\rightarrow$ aTb, str cannot be deriv w/G

l) T T $\rightarrow$ XX, str can be deriv w/G

m) T T $\rightarrow$ XXX, str can be deriv w/G

n) F S $\rightarrow$ E str cannot be deriv w/G

False = F	string = str
True = T	deriv = derived

o) L(G)

Consists of str's that are NOT palindromes & are formed over terminal symbols A & B

2.4 Give context-free grammars that generate the following languages. In all parts, the alphabet Σ is $\{0,1\}$.

- ^A**a.** $\{w \mid w \text{ contains at least three 1s}\}$
- b.** $\{w \mid w \text{ starts and ends with the same symbol}\}$
- c.** $\{w \mid \text{the length of } w \text{ is odd}\}$
- ^A**d.** $\{w \mid \text{the length of } w \text{ is odd and its middle symbol is a 0}\}$
- e.** $\{w \mid w = w^{\mathcal{R}}, \text{ that is, } w \text{ is a palindrome}\}$
- f.** The empty set

2.4)

sym = symbol L^* language
 gene = generate

A) $\Sigma \in \{0,1\}$

The CFG that gene L

$\{w | w \text{ contains at least 1 '3'}\}$ is $S \rightarrow P1P1P1P$ ✓

$R \rightarrow 0P11P1E$

b) CFG that gene L $\{w | w \text{ starts \& ends w/ the same sym.}\}$

$S \rightarrow 0P01P1011$ $P \rightarrow 0P11P1E$

c) $\{w | \text{the length of } w \text{ is odd}\}$

$S \rightarrow 011050105114501151$ OR

$\rightarrow 0111005101511051115$

d) $S \rightarrow 01050105114501151$

e) $S \rightarrow 0110501151/E$ for $\{w | w = w^R\}$ that is, w is a palindrome

f) The empty set CFG is $\{ \}$ which is $S \rightarrow S$

D.C 2.4:

✓

$S \rightarrow 011100511051115$ (or)

$S \rightarrow 0111050105111501151$

d) The CFG that gene the lang $\{w | \text{the length of } w \text{ is odd \& it's mid sym is a } 03\}$ is

$S \rightarrow 01050105111501151$

~~e)~~ ✓

2.13 Let $G = (V, \Sigma, R, S)$ be the following grammar. $V = \{S, T, U\}$; $\Sigma = \{0, \#\}$; and R is the set of rules:

$$\begin{aligned} S &\rightarrow TT \mid U \\ T &\rightarrow 0T \mid T0 \mid \# \\ U &\rightarrow 0U00 \mid \# \end{aligned}$$

a. Describe $L(G)$ in English.



273a)

$S \rightarrow TT|U$

$T \rightarrow 011TO1\#$

$U \rightarrow 0U00\#$

$G = (V, \Sigma, R, S)$

$V = \{S, T, U\}$

$\Sigma = \{0, 1, \#\}$

a) strings containing exactly 2 #'s & any # of 0's

long explanation derive string gene by L

$S \rightarrow TT \rightarrow \#T \rightarrow \#\#$

$S \rightarrow U \rightarrow \# \quad S \rightarrow TT \rightarrow OTT \rightarrow 0\#\#$

$S \rightarrow TT \rightarrow TOT \rightarrow 00\#T \rightarrow 00\#\#$

etc also for U

So Every string containing 2 #'s & any no of zeros.

So the S of the form $0^* \# 0^* \# 0^*$ & $0^n \# 0^n$ are accepted by lang.

$S \rightarrow U \rightarrow 0U00 \rightarrow 0\#00$

str formed by these rules:

D.C The lang $\langle G \rangle$ generated by the given grammar consists of
 • str can consist of 2 pts of the form generated by T concat together, or directly follow the form gener by U.

The T non-terminal generates str that contain any # of 0's followed by the special symbol '#', or a sequence when 0's & #'s alternate.

The U non-term generates str of the form 0^n followed by an even # of 0's, or a single 'X' sym.

Therefore
 The lang $\langle G \rangle$ consists of str that are either
 1. 2 pts that each consist of a sequence of alternating 0's & a single '#' at the end.

2. str that begin w/ '0' & contain an even # of 0's, or just the '#' symbol.

EX: $0\#, 00\#, 0U00, \#, \text{etc.}$

2.6 Give context-free grammars generating the following languages.

a. The set of strings over the alphabet $\{a,b\}$ with more a's than b's

b. The complement of the language $\{a^n b^n \mid n \geq 0\}$

[REDACTED]

[REDACTED]

2.6A) the context-free grammar generating the lang.

$$S \rightarrow Aa|Bs|SBA$$

$$A \rightarrow Aa|E$$

$$B \rightarrow E|BB|bBa|abb$$

So S will generate all str. w/ as many a's & b's R. Forces are extra a which gives the req. str. of the lang.

2.6B) Let L be lang that's comp of given lang L is obtained by
 $L = \{a^n b^m : n \neq m\} \cup \{a^n b^m : n \neq m\}$ Let us say.
 $L = \{a^n b^m : n \neq m\} \cup \{a^n b^m : n \neq m\}$ The free gram

$$S_1 \rightarrow aS_1 b | T | U$$

$$T \rightarrow a | a$$

$$U \rightarrow U b | a$$

$$L_2 =$$

$$S_2 \rightarrow R b a R$$

$$R \rightarrow R R | a | b | E$$

Thus Free gram for L is...

$$L = L_1 \cup L_2$$

$$S \rightarrow S_1 | S_2$$

$$S_1 \rightarrow aS_1 b | T | U$$

$$S_2 \rightarrow R b a R$$

$$T \rightarrow a | a$$

$$U \rightarrow U b | a$$

$$R \rightarrow R R | a | b | E$$

2.6a D.C:

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The Context-Free grammar gener. the given lang

$$S \rightarrow Aa \mid BS \mid SBA$$

$$A \rightarrow Aa \mid \epsilon$$

$$B \rightarrow \epsilon \mid BB \mid bBa \mid aBb$$

gram = grammar
gener = generate

in the above gram. S will gener all str's w/as many a's as b's R Forces are extra on which gives the new str's of the lang.

2.6b D.C: The complement of the lang $\{a^n b^n : n \geq 0\}$ let L be the lang that is a compl of given lang. L can be obtained as $L = \{a^n b^m : n \neq m\} \cup \{a^n b^m : n \neq m\}$

$$L_1 = \{a^n b^m : n \neq m\} \quad L_2 = \{a^n b^m : n \neq m\}$$

The context-free gram gener $L_1 =$

$$S \rightarrow aS_1 b \mid T \mid U$$

$$T \rightarrow aT \mid a$$

$$U = U b \mid a$$

The Context-Free gram gener $L_2 =$

$$S_2 \rightarrow R b a R$$

$$R \rightarrow R R \mid a \mid b \mid \epsilon$$

Thus, the context all together to gener lang L =

$$L = L_1 \cup L_2$$

$$S \rightarrow S_1 \mid S_2$$

$$S_1 \rightarrow aS_1 b \mid T \mid U$$

$$S_2 \rightarrow R b a R$$

$$T \rightarrow aT \mid a$$

$$U \rightarrow U b \mid a$$

$$R \rightarrow R R \mid a \mid b \mid \epsilon$$

2.9 Give a context-free grammar that generates the language

$$A = \{a^i b^j c^k \mid i = j \text{ or } j = k \text{ where } i, j, k \geq 0\}.$$

Is your grammar ambiguous? Why or why not?

2.9) $A = \{a^i b^j c^k \mid i = j \text{ or } j = k \text{ where } i, j, k \geq 0\}$

$\begin{matrix} & i & j & k & \\ & \swarrow & \downarrow & \searrow & \\ i \geq 0 & j \geq 0 & k \geq 0 \end{matrix}$

Lang A is divided in 2 lang which are:

$$A_1 = \{a^i b^j c^k \mid i, j, k \geq 0, i = j\}$$

$$A_2 = \{a^i b^j c^k \mid i, j, k \geq 0, j = k\}$$

getting these lang A_1 & A_2 we can do CFG. Gram of LA is union

$$S \rightarrow S_1 \mid S_2$$

Lang A_1 , the V of I & J = of A's & B's CGFA =

$$S_1 \rightarrow S_1 C \mid E \mid E \rightarrow a E \mid E$$

$$CGFA =$$

$$S_2 \rightarrow a S_2 \mid F \mid E \quad F \rightarrow b F \mid E$$

We generated a string $w = a^i b^j c^k$

w/ Lang A either S_1 or S_2 can be utilized.

Thus CFG of the lang's A = ambiguous

CFL Closure: 2.16

2.16 Show that the class of context-free languages is closed under the regular operations, union, concatenation, and star.

2.16] CFL is Closed under union op, we say $\rightarrow$ 2 start var. s_1, s_2
 for 2 diff lang L_1, L_2
 $S \rightarrow s_1 | s_2$
 if they CFL then $\cup$ of both lang should belong to CFL thus $L_1 \cup L_2 \in CFL$ thus CFL is closed w/ $\cup$ op.
Concat

$S \rightarrow s_1 s_2$

$\forall L_1, L_2 \in CFL$

$\exists w_1, w_2 : w_1 \in L_1, w_2 \in L_2 \Rightarrow w_1 w_2 \in CFL$

thus, w Concat $\in CFL$

$\exists w_1, w_2 : w_1 \in L_1, w_2 \in L_2 \Rightarrow w_1 w_2 \in CFL$ thus, CFL is closed w/ Concat Op.

star: op

1 variable s_1 for lang L_1
 $S \rightarrow s_1^*$

$\forall L_1 \in CFL$ also closed.

~~Star~~ s_1, s_2 for 2 diff lang L_1, L_2 .

$S \rightarrow s_1 | s_2$

if both lang belongs to CFLang then union of both the lang should belong to CFLang.

by the above let if user generates s_1, s_2 string or both then in that case both lang is generated

$L_1 \cup L_2 \in CFL$

CFLang is closed under union operation.

~~in order to show~~ in order to show that CFLang is closed under concat operation
considers 2 start variable S_1 & S_2 for 2 differ Lang L_1 & L_2
Grammar for union oper = $S \rightarrow S_1 S_2$
if both Lang belongs to the CFL then concat of both the
Lang should belong to Context Free Lang.
 $\forall L_1, L_2 \in CFL$

$\{W, W_2 : W \in L_1 \wedge W_2 \in L_2\} \in CFL$

This implies context Free Lang is closed under concat oper.

CFLang closed under star operation

Consider one star var S_1 for the Lang L_1 .
Gram for union operation is as shown:

$S \rightarrow S_1 S_1^*$

if the Lang belongs to the CFLang then star of the
Lang should belong to the CFLang.

$\forall L \in CFL$

Implies CFLang is closed under star oper